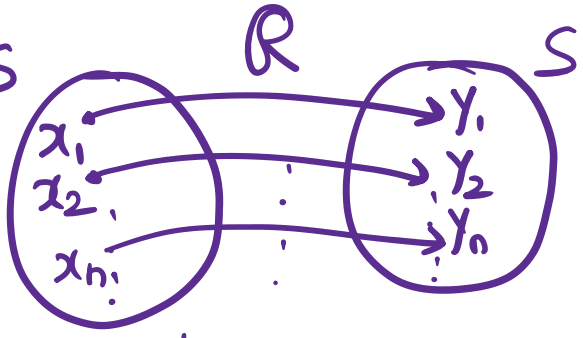
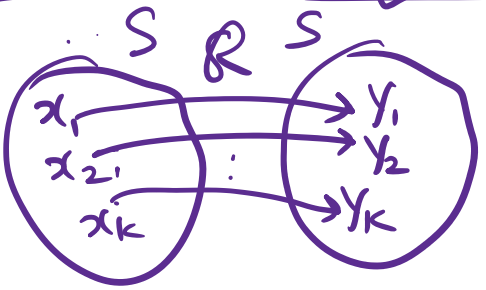
System R :

$y = \log x$, $y = x^2$



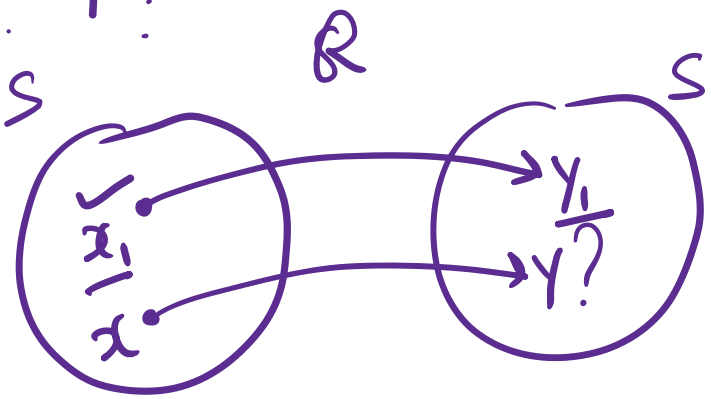


Worst

x	y
x_1	y_1
x_2	y_2
x_n	y_n

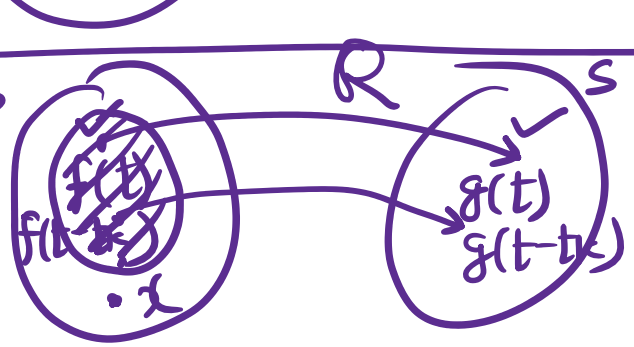
x	y
x_1	y_1
x_2	y_2
x_k	y_k

Better



Best

LTI



$$\sum_{k=-\infty}^{\infty} \alpha_k f(t-t_k) \xrightarrow{R} \sum_{k=-\infty}^{\infty} \alpha_k g(t-t_k)$$

$$x(t) = \sum_{k=-\infty}^{\infty} \alpha_k f(t-t_k) \qquad f(t)?$$

$$x[n] = \sum_{k=-\infty}^{\infty} \alpha_k f[n-k] \quad , f[n]?$$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \quad , f[n] = \delta[n]$$

R , $h[n] = s[n]$

$$f[n]=\delta[n] \xrightarrow[k=-\infty]{R} h[n] = g[n]$$

$$\delta[n] \longrightarrow h[n] \Big\} \text{ Impulse Response of } R$$

$$x[n] = \sum_{k=-\infty}^{\infty} \underbrace{x[k]\,\delta[n-k]}_{x[k]\,h[n-k]} \xrightarrow{R} \underbrace{\sum_{k=-\infty}^{\infty} x[k]\,h[n-k]}_{\triangleq\,(x*h)[n]\ \checkmark}$$

$$= x[n]*h[n] \ \checkmark$$

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau$$

$$\delta(t) \xrightarrow{R} h(t) \longrightarrow \text{impulse response}$$

$$\Rightarrow \quad x(t) \xrightarrow{R} y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$$

$$\triangleq (x*h)(t) \Big\} \text{ Convolut}^n$$

$$y(t) = x(t)*h(t)$$

(I) $x*h = h*x \longrightarrow \int_{-\infty}^{\infty} h(\tau)\,x(t-\tau)\,d\tau$ $\therefore$ Comm. ✓

$$\int_{-\infty}^{\infty} x(\tau)\,h(\underbrace{t-\tau}_{\tau'=t-\tau})\,d\tau = \int_{-\infty}^{\infty} h(\tau)\,x(t-\tau)\,d\tau$$

(II) $x*(h_1+h_2) = x*h_1 + x*h_2$ ✓

(III) $(x*h_1)*h_2 = x*(h_1*h_2)$ ✓ Assoc. ✓